

Conclusion: Learned about different kinds of
topologies & network commands.

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7. default gateway for all adapters.

5. Net Config: Using net config command we can configure server & workstation services, we can configure few settings using this command.
4. Netstat: Derived from the words network & stat. It delivers basic statistics on all network & informs when on which port and addresses the connections, (TCP, UDP) are running & which ports are open for task.
5. Tracert / tracert name command or tracert command. The tracert command is a Command Prompt command used to show several details about the path that a packet takes from computer or device you're on whatever destination you specify.
6. Nslookup: Stands for Name Server Lookup. It is a useful command for getting information from DNS server. It is a network administration tool for the Domain Name System (DNS) to obtain domain name or IP address mapping or any other specific record. It is also used to troubleshoot DNS related problems. To use in interactive mode net type nslookup at the command line & hit return.

Advantages of Hybrid technology

- We can choose the topology based on the requirement for ex, reliability is our concern then we can use star topology instead of bus topology.
- Scalable as we can connect other computer networks with the existing network issues with different topologies.

Disadvantages of Hybrid technology.

- Fault detection is difficult.
- Installation is difficult.
- Design is complex so maintenance is high thus expensive.

Networking commands.

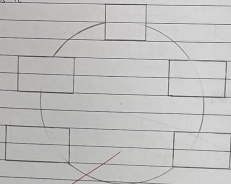
1. Ping: The ping command is a command prompt command used to test the ability of the source computer to reach a specified destination. It's usually used as a simple way to verify that a computer can communicate over networks with another computer or network device.
2. Ipconfig: IPConfig stands for Internet Protocol Configuration. This is a command line operation which displays all the current TCP/IP (Transmission Control Protocol/Internet Protocol) network configuration regardless the DHCP (Dynamic Host Protocol) & DNS (Domain Name Service). It also displays IP address, subnet mask.

4. Ring Topology

In ring topology each device is connected with the devices on either side of it. There are two dedicated point-to-point links a device has with the devices on either side of it.

The structure forms a ring thus it is known as ring topology.

If a device wants to send data to another device then it sends a data in one direction, each device in it sends the data. The ring topology has a repeater if the received data is intended for another device, the repeater forwards this data until the intended device receives it.



5. Hybrid Topology

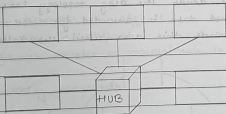
A combination of two or more topologies is known as hybrid topology. For ex, a combination of star mesh topology is known as hybrid topology.

2. Star Topology

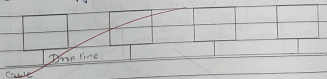
In star topology each device in the network is connected to a central device called hub.

Unlike Mesh topology, star topology doesn't allow direct communication between devices, a device must have to communicate through hub.

If one device wants to send data to other device, it has to first send the data to hub, then the hub transmits that data to the designated device.



3) Bus Topology



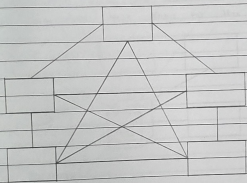
In bus topology there is a main cable, all the devices are connected to this main cable through drop lines.

There is a device called tap that connects the drop lines to main cable. Since all the data is transmitted over the main cable, there is a limit of drop lines, the distance a main cable can have.

Computer Network Technology.

Geometric representation of how the computers ^{are} connected to each other is known as topology. There are five types of topology - Mesh, Star, Bus, Ring & Hybrid.

1. Mesh topology.



In mesh topology each device is connected to every other device on the network through a dedicated point-to-point link.

When we say dedicated it means that the link only carries data for the two connected devices only. Let's say we have 'n' devices in the network then each device must be connected with $(n-1)$ devices of network.

- Peer-to-peer network has no dedicated server.
- Special permissions are assigned to each computer for sharing the resources, but this can lead to a problem if the computer with that resource is down.

Client / Server Network:

- Client / Server network is a network model designed for the end users called clients, to access the resources such as songs, videos, etc. from a central computer known as server.
- The central controller is known as a server and all other computers known as clients.
- The central controller is known as a server. A server performs all the major operations such as network management.
- A server is responsible for managing all the resources such as files, directories, printers, etc.
- All the clients communicate with each other through server. For ex: if client 1 wants to send some data to client 2, then it first sends the request to the server for the permission. The server sends the response to client 1 to initiate its communication with the client 2.

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Aim : To perform network commands & Draw Network architecture.

Theory : Computer Network Architecture is defined as the physical & logical design of the software, hardware, protocols & media of the transmission of data.

The two types of network architectures are used :

Types of Computer Network

Peer-to-peer
network

Client/Server
network

Peer-to-peer Network :

- Peer-to-peer network is a network in which all the computers are linked together with equal privilege & responsibilities the data.
- Peer-to-peer network is useful for small environment usually up to 10 computers.